

2D Projectile Motion: Video Tracking & Kinematics Analysis

AP Physics C: Mechanics • Laboratory Investigation • Platform: Lab-ove and Beyond by MR. F.

Course: AP Physics C (Mechanics)	Period / Block: _____	Class Duration: 42 Minutes
Student Name: _____	Lab Partner: _____	Date: _____

Objective: Track 2D projectile motion in Lab-ove and Beyond. Calibrate scale, measure kinematics parameters, fit linear and quadratic curves to extract g , and apply calculus to analyze trajectory extrema.

■ AP PHYSICS C: UNIFORMLY ACCELERATED MOTION (UAM) REFERENCE FORMULAS

$$(1) v = v_0 + a \cdot t \quad (3) v^2 = v_0^2 + 2a \cdot \Delta x$$

$$(2) \Delta x = v_0 \cdot t + \frac{1}{2}a \cdot t^2 \quad (4) \Delta x = \left(\frac{v_0 + v}{2}\right) \cdot t$$

* Note: Apply component-wise with $a_x = 0$ (horizontal) and $a_y = -g$ (vertical). No further equations are provided.

Your 42-Minute Clock

The five steps below add up to 42 minutes (17 of 25 points). Section 4 is an optional homework extension (8 points).

Class Time	Lab Phase	What You Actually Do	Pts
0:00 - 0:04 (4m)	1. Set Up & Calibrate	Load 'FizziQ: Parabolic Motion (Classic Lab)' (30.0 FPS, $\Delta t = 0.0333$ s). Ruler tool: click both ends of the 1.00 m rod and enter 1.00 m. Origin tool: click the ball at Frame 0, then set +Y up and +X downrange.	—
0:04 - 0:14 (10m)	2. Track & Record	Autotrack () the ball across the 25 tracked frames, then export the CSV. Copy x, y, vx, vy and v into the 5-milestone data table.	1
0:14 - 0:25 (11m)	3. Physical Measurements	Section 2 - Q2.1 launch speed, angle and vector (3 pts). Q2.2 derive & predict apex/range, then measure and verify (3 pts). Q2.3 mid-ascent vectors, direction angle and $a_{ }$ / $a_{\perp}$ (3 pts).	9
0:25 - 0:35 (10m)	4. Graphs & Curve Fits	Section 3 - x(t) linear fit and a_x (2 pts). y(t) quadratic fit, g_{exp} and % error (3 pts). vy(t) fit comparison and the y(x) trajectory shape (2 pts).	7
0:35 - 0:42 (7m)	5. Check & Submit	Verify every answer carries SI units and the correct sign. Compare your fitted values with the app readouts. Staple the handout and submit.	—
42 minutes	CORE TOTAL	Required work that is graded during the period.	17
+25 min outside class (25m)	6. Optional Extension	Section 4 - AP Physics C calculus challenges: C4.1 trajectory derivation (3 pts), C4.2 position integration & impact velocity vector (3 pts), C4.3 minimum-speed extremum (2 pts). Not required to finish the 42-minute lab.	8

■ HOW TO SPEND THE PERIOD

- REQUIRED items are the 42-minute core: Steps 1-5 and Sections 2-3. Finish and submit these today.
 - ◇ OPTIONAL EXTENSION items are Section 4 (Challenges 4.1-4.3). They need about 25 more minutes as homework; nothing in Section 4 is needed to finish class today.
- Pace yourself: if the clock reaches 0:25, you should already be starting Section 2.

1. What to Do on the 'Lab-ove and Beyond' App

■ APP WORKFLOW PROTOCOL

1. Load Video: Top navbar → 'Video Library' → select 'FizziQ: Parabolic Motion (Classic Lab)' (30.0 FPS, $\Delta t = 0.0333$ s).
2. Calibrate Scale (📏 Ruler Tool): Click Ruler tool. Click bottom and top of the 1.00 m rod on left → Enter '1.00 m'.
3. Set Origin (📍 Axes Tool): Click Origin tool. Click projectile center at Frame 0 (release point). Set +Y Up and +X Downrange.
4. Track Motion (+ Crosshair / ⚡ Autotracker): Click projectile center across all 25 frames from release ($t = 0.00$ s) to landing.
5. Vectors (🕒): Toggle Velocity and Acceleration overlays to inspect vector magnitude and direction live.

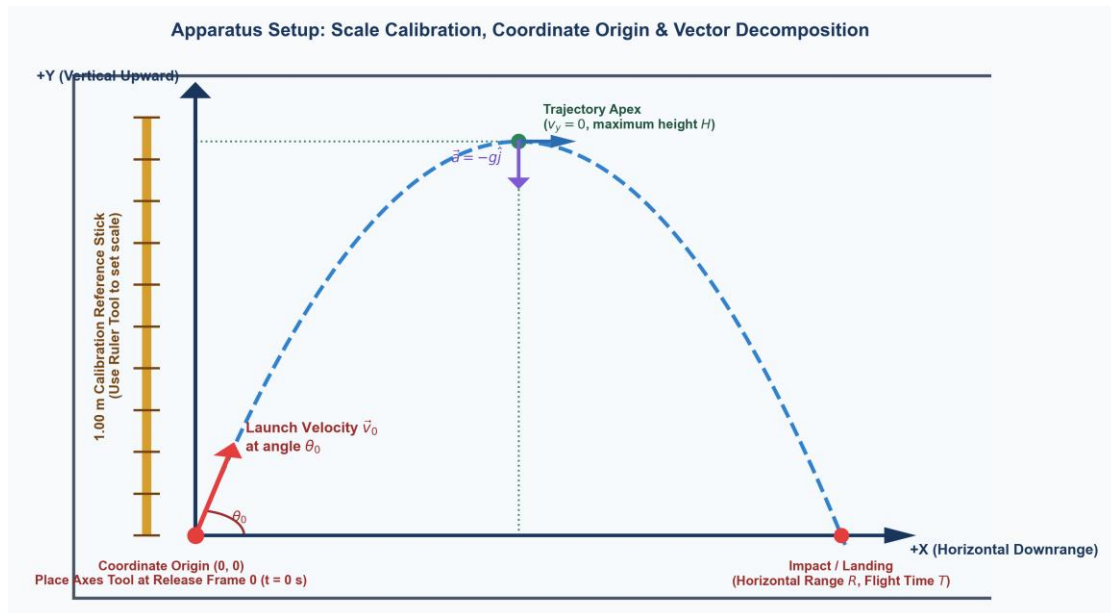


Figure 1: Lab-ove and Beyond experimental interface: 1.00 m calibration stick, launch origin (0,0), and coordinate axes.

2. Physical Measurements (Data Acquisition)

Read instantaneous values by hovering on tracked points or opening the Table view. Record the 5 key milestones:

Trajectory Milestone	Frame	Time t (s)	x (m)	y (m)	v_x (m/s)	v_y (m/s)	Speed v (m/s)
1. Launch / Release	0	0.000	0.000	0.000			
2. Mid-Ascent	6	0.200					
3. Apex (Peak Height)	13	0.433				≈ 0.00	
4. Mid-Descent	18	0.600					
5. Landing / Impact	24	0.800					

● **REQUIRED** — counted in the 42-minute core | budget about 4 min | 3 pts

Question 2.1 [Launch Velocity & Angle — 3 Points]:

(a) Initial Launch Speed [1 Pt]: Using your measured launch velocity components at Frame 0 ($t = 0.00$ s), determine the initial launch speed v_0 .

✍ Show formula, values, and calculated v_0 in m/s:

(b) Launch Angle [1 Pt]: Determine the launch angle θ_0 above the horizontal.

✍ Show formula, values, and calculated θ_0 in degrees:

(c) Launch Vector [1 Pt]: Express the launch velocity vector $\vec{v}(0)$ in Cartesian unit-vector notation ($\hat{i}$, $\hat{j}$).

✍ Write $v(0)$ with numerical components and units:

• **REQUIRED** — counted in the 42-minute core | budget about 4 min | 3 pts

Question 2.2 [Derive First & Predict Extrema, Then Measure & Verify — 3 Points]:

(a) Apex Time & Maximum Height: Algebraic Derivation & Prediction [1 Pt]: Starting from the 4 UAM equations at the top, algebraically derive symbolic formulas for apex time t_{apex} and maximum height H in terms of vertical launch component v_{0y} and g . Substitute your measured launch component and $g = 9.81 \text{ m/s}^2$ to predict their numerical values before checking your video data.

Show algebraic UAM derivations for t_{apex} and H , then calculate predicted values:

(b) Total Flight Time & Range: Algebraic Derivation & Prediction [1 Pt]: Starting from the 4 UAM equations, algebraically derive symbolic formulas for total flight duration T and horizontal range R on level ground in terms of launch velocity components and g . Calculate your theoretical predictions.

Show algebraic UAM derivations for T and R , then calculate predicted values:

(c) Video Data Measurement & Verification [1 Pt]: Now inspect your tracked data in Lab-ove and Beyond. Read and record measured H and t_{apex} at Frame 13, and measured R and T at Frame 24. Calculate the percent error between your theoretical predictions and video measurements. Does the flight exhibit symmetry?

State measured values (Frames 13 & 24), compute % errors, and evaluate symmetry:

● **REQUIRED** — counted in the 42-minute core | budget about 3 min | 3 pts

Question 2.3 [Instantaneous Vector Kinematics at Mid-Ascent ($t = 0.200$ s) — 3 Points]:

(a) Position & Velocity Vectors [1 Pt]: At $t = 0.200$ s (Frame 6), state position vector $\vec{r}(0.20)$ and velocity vector $\vec{v}(0.20)$ in unit-vector notation.

✍ Write $r(0.20)$ and $v(0.20)$ in unit-vector notation:

(b) Direction of Motion [1 Pt]: Determine the instantaneous direction angle ϕ of travel above the horizontal at this instant.

✍ Show formula and calculate direction angle ϕ in degrees:

(c) Acceleration Components [1 Pt]: Decompose the acceleration due to gravity into its tangential component $a_{\parallel}$ and normal component $a_{\perp}$. Explain the physical effect of each component on the projectile's motion.

✍ Derive and compute a_{parallel} and a_{perp} (m/s^2) with physical explanations:

3. How to Draw & Fit Graphs in the App

1. Open Graphs: Switch to 'Graphs' tab (or Split View).
2. Select Variables: Plot 'x', 'y', 'vy', and 'Trajectory y(x)'.
3. Apply Fits: Click 'Curve Fit' () → choose 'Linear Fit' or 'Quadratic Fit'.

• **REQUIRED** — counted in the 42-minute core | budget about 3 min | 2 pts

Question 3.1 [Graph 1: Horizontal Motion x(t) Linear Fit — 2 Points]:

(a) Linear Regression Equation & R^2 [1 Pt]: Plot x(t), apply a linear fit, and record your regression equation and R^2 value.

✍ State x(t) regression equation and R^2 value:

(b) Slope Interpretation & Acceleration [1 Pt]: What physical quantity does the slope represent? What does this result prove about horizontal acceleration a_x ? Justify using calculus.

✍ Explain slope meaning and deduce a_x using calculus:

• **REQUIRED** — counted in the 42-minute core | budget about 4 min | 3 pts

Question 3.2 [Graph 2: Vertical Motion y(t) Quadratic Fit & g Extraction — 3 Points]:

(a) Quadratic Regression Equation & R^2 [1 Pt]: Plot y(t), apply a quadratic fit ($y = At^2 + Bt + C$), and record your regression equation and R^2 value.

✍ State y(t) quadratic regression equation and R^2 value:

(b) Gravitational Acceleration Extraction [1 Pt]: Using your quadratic fit and UAM kinematics, determine your experimental acceleration due to gravity g_{exp} . Show your work and reasoning.

✍ Show derivation and calculate g_{exp} in m/s^2 :

(c) Percentage Error [1 Pt]: Calculate the percentage error of your experimental value relative to standard $g = 9.81 m/s^2$.

✍ Calculate percentage error:

● **REQUIRED** — counted in the 42-minute core | budget about 3 min | 2 pts

Question 3.3 [Graph 3 & 4: Velocity $v_y(t)$ Fit & Trajectory $y(x)$ — 2 Points]:

(a) Vertical Velocity Fit [1 Pt]: Plot $v_y(t)$, apply a linear fit, and compare your experimental slope and vertical intercept to theoretical expectations.

✍ State $v_y(t)$ equation and compare slope and intercept to theory:

(b) Trajectory Shape & Air Drag [1 Pt]: Plot the spatial trajectory $y(x)$. Describe the shape and evaluate whether real-world aerodynamic drag causes any observable deviation near landing.

✍ Describe 2D trajectory shape and note any real-world drag deviations:

◇ **OPTIONAL EXTENSION** — not needed to finish the 42-minute lab | budget about 25 min | 8 pts

4. Optional Extension: AP Physics C Free-Response Calculus Challenges

◇ **OPTIONAL EXTENSION** — not needed to finish the 42-minute lab | budget about 8 min | 3 pts

Challenge 4.1 [Calculus Trajectory Derivation & Path Verification — 3 Points]:

(a) Parametric Trajectory Derivation [1 Pt]: Starting from the horizontal and vertical position functions $x(t)$ and $y(t)$, eliminate time t to derive the symbolic trajectory equation $y(x)$ in terms of v_{0x} , v_{0y} , and g .

✍ Show complete algebraic steps to eliminate time t and derive $y(x)$:

(b) Numerical Trajectory Equation [1 Pt]: Using your experimental launch data, evaluate the coefficients and write the numerical equation for your trajectory $y(x)$.

✍ Write numerical equation $y(x)$ with evaluated coefficients:

(c) Apex Condition via Calculus [1 Pt]: Using calculus, determine the horizontal coordinate of the apex x_{apex} by finding the extremum of $y(x)$. Calculate its value and compare to your data table.

✍ Apply derivative condition to find x_{apex} and compare to data table:

◇ **OPTIONAL EXTENSION** — not needed to finish the 42-minute lab | budget about 9 min | 3 pts

Challenge 4.2 [Calculus Definite Integration: Vertical Position Integral & Impact Vector — 3 Points]:

(a) Position Integral & Maximum Height [1 Pt]: Starting from the vertical velocity function $v_y(t) = v_{0y} - gt$, use definite calculus integration $y(t) = \int v_y(t') dt'$ to derive $y(t)$. Prove that evaluating the integral over $[0, t_{apex}]$ rigorously yields the maximum height formula.

✍ Show definite calculus integration steps to find $y(t)$ and prove H :

(b) Impact Velocity Vector [1 Pt]: At landing time T , evaluate velocity components $v_x(T)$ and $v_y(T)$. Express the landing velocity vector $\vec{v}(T)$ in Cartesian unit-vector notation ($\hat{i}, \hat{j}$).

✍ Evaluate landing components and write $v(T)$ in unit-vector notation:

(c) Impact Angle & Kinetic Energy Conservation [1 Pt]: Determine the impact angle θ_{impact} below the horizontal. Compare impact speed $|v(T)|$ to launch speed v_0 . Justify why speed is conserved using the Work-Energy Theorem.

✍ Calculate θ_{impact} , compare $|v(T)|$ vs v_0 , and justify with Work-Energy Theorem:

◇ **OPTIONAL EXTENSION** — not needed to finish the 42-minute lab | budget about 8 min | 2 pts

Challenge 4.3 [Calculus Extremum: Minimum Speed During 2D Flight — 2 Points]:

(a) Speed Optimization [1 Pt]: Express the square of the projectile speed $v^2(t)$ as a function of time. Using calculus, determine the exact time t^* at which the speed reaches a minimum.

✍ Formulate $v^2(t)$, differentiate with respect to t , and solve for t^* :

(b) Minimum Speed Value & Verification [1 Pt]: Determine the magnitude of the minimum speed v_{min} and explain its physical significance. Compare with your experimental speed measurement at the apex.

✍ State v_{min} , explain physical meaning, and compare to apex data:

5. Student Evaluation Rubric (25 Points = 17 Core + 8 Extension)

Section	Performance Criteria	Max Pts	Score
CORE · App Setup & Data Table	Video loaded, 1.00 m rod calibrated, origin set at Frame 0, 5 milestones recorded	1	
CORE · Section 2: Physical Measurements	Q2.1 launch speed/angle/vector (3), Q2.2 derive & predict apex/range, then verify (3), Q2.3 mid-ascent vectors (3)	9	
CORE · Section 3: App Graphing & Fits	Q3.1 $x(t)$ fit & a_x (2), Q3.2 $y(t)$ fit & g extraction (3), Q3.3 $v_y(t)$ & trajectory (2)	7	
CORE SUBTOTAL	Everything above is required and due at the end of the 42-minute period	17	
EXTENSION · Section 4: Calculus Challenges	Optional homework, not needed to finish the lab: C4.1 (3), C4.2 position integral & impact (3), C4.3 (2)	8	
TOTAL SCORE	AP Physics C Mechanics Laboratory Investigation Grade	25	